AI-Based Question Answering System Using a Small Knowledge Dataset.

➢ Problem Statement:

Students and users often need quick answers to questions about an organization or institution. Searching through large amounts of information manually can be time-consuming. The proposed AI system will answer user questions using a small, structured question-answer dataset. The system will identify the most relevant information from the dataset and provide the corresponding answer.

➢ Target User:

The main users of this system are students, applicants, visitors, and anyone who wants to get quick and simple information about the institution. The system will help them find answers to their questions without having to search through a lot of information manually.

➢ AI Use Case:

Task Type: Question Answering (Q&A):

The main idea of this AI system is to allow users to ask questions in their own words and get the most relevant answer from the given dataset. This will make it easier for users to find the information they need without searching through the entire dataset.

Example 1:

User question:
What is SPIT?

Expected answer:
SPIT stands for Sardar Patel Institute of Technology.

Example 2:

User question:
What does SPIT stand for?

Expected answer:
SPIT stands for Sardar Patel Institute of Technology.

➢ Data Source:

For this project, I am using the task1dataset.csv file as the main source of information. The dataset contains 500 question-and-answer pairs. It has two columns:

- **Question** – contains the questions that users may ask.
- **Answer** – contains the answers related to those questions.

This dataset will be used as the knowledge base for the Q&A system, helping it find the most relevant answer to a user's question.

➢ Proposed AI Approach:

The system will take a question from the user and look through the dataset to find the most similar question. It will then use the matching question to provide the corresponding answer to the user. This will help the system give relevant answers even when the user asks a question in a slightly different way.

➢ Constraints:

The system will mainly use the information available in the given dataset to answer questions. Since the dataset contains only 500 question-and-answer pairs, it may not be able to answer questions that are outside the information provided. The system should also avoid giving unrelated answers when it cannot find a suitable match for the user's question.

➢ Success Criteria:

Metric	Success Target
Answer accuracy	At least 90%
Relevant answer retrieval	At least 90%
Incorrect or unrelated answers	Less than 10%
Response coverage	Correctly answer most questions related to the dataset

- The system will be considered successful if it can provide the correct or relevant answer for at least 90% of the test questions. It should also avoid giving unrelated answers when it cannot find a suitable match.

➢ Evaluation Approach:

To evaluate the system, some questions from the dataset will be used for testing. The answers provided by the system will then be compared with the expected answers in the dataset.

The evaluation will mainly check:

1. Whether the system provides the correct answer.
2. Whether it can understand questions even when they are worded differently.
3. Whether the answer is relevant to the user's question.
4. Whether the system avoids giving unrelated answers.

The main target will be to achieve at least **90% correct answers** on the test questions.

➢ Example Input and Output:

User Input	Expected Output
What is SPIT?	SPIT stands for Sardar Patel Institute of Technology.
What does SPIT stand for?	SPIT stands for Sardar Patel Institute of Technology.
What is the full name of SPIT?	Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology.

➢ Final Objective:

The aim of this project is to create a simple AI-based Q&A system that can find relevant information from the given dataset and provide useful answers to users. The project will mainly focus on checking how accurately the system can understand user questions and find the appropriate answers from the available data.